

A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

John A. Janik
john.janik@gmail.com

June 2026

Abstract

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length $M = 240$ over the alphabet $\{1, 2, 4\}$, calibrated to mean $\log_2 3$ within $2/M$ and carrying a non-coboundary order-3 shadow bias, sampled from the law $p_1 = 0.567$, $p_2 = 0.357$, $p_4 = 0.076$. Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo 2^{B_M+1} has least positive representative of nearly full bit length, not stabilizing between prefix lengths $M/2$ and M — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope $\alpha = 2 - \log_2 3$ (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of $\mathbb{Z}_2$; the Collatz conjecture asserts that classes like these never meet $\mathbb{Z}^+$.

1 Conventions

For a valuation word $w = (b_0, \dots, b_{M-1})$, $B_N = \sum_{n < N} b_n$; the set of odd x whose first N valuations equal the prefix is a single residue class modulo 2^{B_N+1} ; “rep bits” is the bit length of its least positive representative r_N , “mod bits” is $B_N + 1$, and the fill ratio is their quotient. A positive realization would force r_N to stabilize at $\lceil \log_2 x_0 \rceil$ once $2^{B_N} > x_0$; a ghost rides the diagonal $r_N \sim 2^{B_N}$. The bias is $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$, the order-3 non-coboundary two-point shadow.

2 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios ≈ 1 .

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
1	1.5833	0.214	225	380	381	381	1.000
2	1.5792	0.214	225	379	379	380	0.997
3	1.5792	0.214	225	379	378	380	0.995
4	1.5833	0.239	205	380	381	381	1.000
5	1.5917	0.220	213	382	383	383	1.000
6	1.5792	0.215	225	379	379	380	0.997
7	1.5917	0.215	225	382	383	383	1.000
8	1.5792	0.216	225	379	380	380	1.000
9	1.5792	0.229	214	379	380	380	1.000
10	1.5792	0.233	225	379	380	380	1.000
11	1.5792	0.229	225	379	379	380	0.997
12	1.5792	0.216	225	379	380	380	1.000
13	1.5917	0.233	193	382	382	383	0.997

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
14	1.5792	0.215	224	379	380	380	1.000
15	1.5917	0.216	208	382	383	383	1.000
16	1.5833	0.214	225	380	378	381	0.992
17	1.5917	0.215	225	382	380	383	0.992
18	1.5792	0.218	225	379	380	380	1.000
19	1.5917	0.220	225	382	383	383	1.000
20	1.5792	0.260	214	379	380	380	1.000
21	1.5792	0.272	208	379	380	380	1.000
22	1.5917	0.220	214	382	383	383	1.000
23	1.5875	0.228	225	381	382	382	1.000
24	1.5875	0.246	214	381	380	382	0.995
25	1.5917	0.222	211	382	383	383	1.000
26	1.5917	0.242	189	382	383	383	1.000
27	1.5792	0.222	222	379	376	380	0.989
28	1.5917	0.214	211	382	383	383	1.000
29	1.5875	0.215	225	381	382	382	1.000
30	1.5792	0.260	217	379	375	380	0.987
31	1.5917	0.214	216	382	382	383	0.997
32	1.5833	0.239	203	380	380	381	0.997
33	1.5792	0.218	225	379	377	380	0.992
34	1.5875	0.217	225	381	380	382	0.995
35	1.5917	0.214	224	382	379	383	0.990
36	1.5917	0.216	225	382	377	383	0.984
37	1.5792	0.220	225	379	380	380	1.000
38	1.5917	0.225	225	382	383	383	1.000
39	1.5875	0.221	225	381	381	382	0.997
40	1.5792	0.218	225	379	379	380	0.997
41	1.5917	0.225	206	382	380	383	0.992
42	1.5792	0.238	212	379	377	380	0.992
43	1.5792	0.215	225	379	375	380	0.987
44	1.5917	0.215	225	382	383	383	1.000
45	1.5875	0.217	225	381	379	382	0.992
46	1.5792	0.215	219	379	378	380	0.995
47	1.5792	0.229	223	379	380	380	1.000
48	1.5792	0.215	225	379	377	380	0.992
49	1.5917	0.215	225	382	383	383	1.000
50	1.5917	0.216	225	382	382	383	0.997
51	1.5792	0.229	223	379	377	380	0.992
52	1.5792	0.248	206	379	380	380	1.000
53	1.5917	0.218	224	382	376	383	0.982
54	1.5792	0.215	225	379	380	380	1.000
55	1.5917	0.233	225	382	382	383	0.997
56	1.5917	0.279	194	382	383	383	1.000
57	1.5792	0.215	225	379	380	380	1.000
58	1.5792	0.218	225	379	377	380	0.992
59	1.5792	0.238	201	379	380	380	1.000
60	1.5792	0.214	225	379	380	380	1.000
61	1.5792	0.238	225	379	380	380	1.000
62	1.5833	0.230	213	380	381	381	1.000
63	1.5792	0.222	219	379	379	380	0.997
64	1.5792	0.215	219	379	379	380	0.997
65	1.5792	0.222	225	379	380	380	1.000
66	1.5917	0.216	196	382	383	383	1.000
67	1.5917	0.215	225	382	383	383	1.000

#	mean b	$ D_3 $	$P(16)$	B_{240}	rep bits	mod bits	fill
68	1.5792	0.216	225	379	372	380	0.979
69	1.5917	0.216	225	382	382	383	0.997
70	1.5917	0.216	225	382	383	383	1.000
71	1.5917	0.214	224	382	382	383	0.997
72	1.5792	0.214	225	379	380	380	1.000
73	1.5875	0.221	225	381	379	382	0.992
74	1.5917	0.215	222	382	383	383	1.000
75	1.5792	0.215	225	379	380	380	1.000
76	1.5792	0.225	225	379	380	380	1.000
77	1.5833	0.241	225	380	381	381	1.000
78	1.5917	0.215	225	382	382	383	0.997
79	1.5875	0.221	225	381	380	382	0.995
80	1.5875	0.217	225	381	381	382	0.997
81	1.5917	0.225	220	382	380	383	0.992
82	1.5833	0.214	222	380	381	381	1.000
83	1.5917	0.215	225	382	381	383	0.995
84	1.5917	0.218	225	382	383	383	1.000
85	1.5792	0.222	225	379	378	380	0.995
86	1.5792	0.238	225	379	380	380	1.000
87	1.5792	0.229	225	379	376	380	0.989
88	1.5875	0.227	225	381	382	382	1.000
89	1.5792	0.218	225	379	380	380	1.000
90	1.5917	0.216	225	382	383	383	1.000
91	1.5792	0.216	225	379	378	380	0.995
92	1.5833	0.225	225	380	381	381	1.000
93	1.5917	0.220	220	382	382	383	0.997
94	1.5792	0.214	225	379	380	380	1.000
95	1.5792	0.238	223	379	380	380	1.000
96	1.5792	0.238	225	379	379	380	0.997
97	1.5917	0.233	221	382	383	383	1.000
98	1.5792	0.229	219	379	380	380	1.000
99	1.5917	0.218	223	382	382	383	0.997
100	1.5792	0.215	225	379	379	380	0.997
101	1.5792	0.218	225	379	380	380	1.000
102	1.5792	0.218	216	379	376	380	0.989
103	1.5792	0.229	225	379	379	380	0.997
104	1.5875	0.217	225	381	382	382	1.000
105	1.5917	0.229	211	382	383	383	1.000
106	1.5833	0.232	225	380	380	381	0.997
107	1.5792	0.218	225	379	379	380	0.997
108	1.5792	0.214	225	379	380	380	1.000
109	1.5917	0.220	225	382	381	383	0.995
110	1.5917	0.220	225	382	383	383	1.000
111	1.5792	0.229	225	379	380	380	1.000
112	1.5833	0.218	225	380	379	381	0.995
113	1.5792	0.248	225	379	379	380	0.997
114	1.5792	0.229	223	379	380	380	1.000
115	1.5917	0.220	224	382	383	383	1.000
116	1.5917	0.233	200	382	382	383	0.997
117	1.5792	0.222	225	379	378	380	0.995
118	1.5875	0.214	225	381	382	382	1.000
119	1.5792	0.218	224	379	380	380	1.000
120	1.5792	0.214	225	379	380	380	1.000

3 Sturmian ghosts (critical slope, now a theorem)

Words $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$, $\alpha = 2 - \log_2 3$. By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

θ	$ D_3 $	B_{240}	rep bits	mod bits	fill	rep bits at $N=100/200$
0.000	0.447	380	381	381	1.000	159/316
0.100	0.447	380	380	381	0.997	157/318
0.200	0.447	380	377	381	0.990	159/317
0.330	0.447	380	381	381	1.000	155/318
0.500	0.453	381	379	382	0.992	159/317
0.610	0.453	381	376	382	0.984	160/318
0.750	0.453	381	382	382	1.000	160/317
0.900	0.447	380	381	381	1.000	160/318
0.057	0.447	380	381	381	1.000	159/318
0.085	0.447	380	381	381	1.000	159/318

Example Sturmian words (first 120 letters):

 $\theta = 0.00:$

1221212212122121212212122121212212122121212212122121212
21212212121221212212121221212212122121221212212122121221

 $\theta = 0.10:$

122121212212122121221212212122121212212122121221212212122121221212
1221212212122121212212122121221212212122121221212212122121221212

 $\theta = 0.20:$

1212212122121212212122121221212212122121212212122121221212212
1221212122121221212212122121221212212121212212122121221212212

 $\theta = 0.33:$

12122121212212122121212212122121212212121221212212121221212212
12122121221212122121221212212122121221212212122122121221212212

A The 120 candidate words

Alphabet $\{1, 2, 4\}$, length 240, row-wrapped at 60 letters.

#1 (mean 1.583, $|D_3|=0.214$):

111411211121121111112114421114112222111122212112122212112221
11222112241121212222121212111211111121221222111411112211121
22211111111242222211111121211411212111214221114142121111141
21121141222212111111112222212114121211221411121111114422241

#2 (mean 1.579, $|D_3|=0.214$):

2222222222222211111121121142111121142111122122221114241211
1111124121112114221111411121111221122122121111112121111211
212222111212111114221424111211121411211144212111114121114221
211211111111221211111212112211212111411211412422212222111

#3 (mean 1.579, $|D_3|=0.214$):

2221121111121124212211121112111121212211121111211221144412
42222221112212121111111211111121141212111222422214111212111
112411112112224211112111211122121122111124112111224111121221
224122114112111122122412112421122212111441121112112121111122

#4 (mean 1.583, $|D_3|=0.239$):

[illegible]

#5 (mean 1.592, $|D_3|=0.220$):

[illegible]

221222112412112111422412112222244212121411111124111221221212
212211211411111122112142111122211122224114212211221214211224

#7 (mean 1.592, $|D_3|=0.215$):

```

1211121112112121211112241211121214111212114112411221242222211
221221111112211111221212224214111211122112111222111211112112
122422141112214124211112222141214211211121211121112211111
111141212222421211111221212111121224212121411221222141211

```

[illegible]

#11 (mean 1.579, $|D_3|=0.229$):

```

22222242222224222222211122112212412212211122111422211121222112
111112111211212121112211221212121221221111222412121241141
121211114211121212121211141212121212142221111121124112122212
22422121221221221211114212121124122111111221111112212211111

```

[illegible][illegible]

#17 (mean 1.592, $|D_3|=0.215$):

```

1112112122421212221122122211111212242111211212222114111212
2111112421224121121121211221141122141121212422111222121411
4111111421211221121212112211211221121111212212122121114112
21222111111114114112121411111222112124211121122242421222

```

#19 (mean 1.592, $|D_3|=0.220$):

112411111211112121112411212111212211211111412211111211112221
 12121211112211211212111411111111214112212244212112211222122
 2114142111141121212212244111111212111121222141241114121222
 1111212212244112121112212241121241121242422121221111111

#20 (mean 1.579, $|D_3|=0.260$):
22222242222222222222222222222222222211112211111221124112
21111211221122412111411112221122212122211222111114122212212
421114212124221212112112211111212211111221122124221111211
12121111112122212111121111111111114112111241121222122212111

#21 (mean 1.579, $|D_3|=0.272$):
2222222222222222222222222222222222212121111221221112122222121
11111122121211111222111121111211112221121221114112141421
11212111121121222121212121212122421222111124142121111212121
121212112111112212121221212211222122212441221211111412122

#22 (mean 1.592, $|D_3|=0.220$):
11111111111111111111111111211112242124211142124122111112241
22111421421221122221221414421121122211214122211211112212
1141111121114121211211112122441212112212122242211211111
12111112121212124242212121241212211122411111212122122211212

#23 (mean 1.587, $|D_3|=0.228$):
142222211241411141121112112112241442212211111212121222412
2221211112112212211112112214112121214111112111122122122
1422212121111141214212121222111212111144211124212211121212
12241111111111111121112222211122424411124114214111122111

#24 (mean 1.587, $|D_3|=0.246$):
1111111111111111111111124212211111111111111111122122211412
1212121224142141111111121421221141112112421212222441414212
2141412221211222121121121124111112211421121112122141
212111421211121241244121212212221122214111214214111421112

#25 (mean 1.592, $|D_3|=0.222$):
111111111111111111111111242212211122221224211111411
12112221221124211422112412121221411112122212412221241421
221112412222212122112222424212121242121422222121111122
211111212122122122122121224222121122221111112212212211

#26 (mean 1.592, $|D_3|=0.242$):
1112111111
221222221111221424111224211114111122112144222421212124
21424412112122121411221221411421221114112221122141211
12122244212212212212412211111412422212212141424221221211

#27 (mean 1.579, $|D_3|=0.222$):
224222222222222222222211122211112121121112111112111111
12221122124222222221211111112241121222121111422112141422
4112121211111212122241411221122211111122112121212121
221424122224212211111114111214222211111214222121212121

#28 (mean 1.592, $|D_3|=0.214$):
11111111111111111111111111111211441121411212121112112122
41214121121222122122122222222224212241212122121241211
112121214212121114112211242112411242122222122112122121
121211222111212121442412112244222121211122121141221212

#29 (mean 1.587, $|D_3|=0.215$):
11211241421111111212422111111111121122211141211244211212
42121212221212222121221211221112211142212212214121212124
1121111242212112424121121122211122212112212111212112
1221212121111442114414211212121121421122121212122211111

#30 (mean 1.579, $|D_3|=0.260$):
2222222222222222222222222222222222214111212111112211
22111111211211241122212122414212212211411212121212122
1111112222112121211111111121111114422212121214141221
21221212121122111111212121121222241221211112221222211

#31 (mean 1.592, $|D_3|=0.214$):
11111111111111111111111212122112211221421122122222221214
21222211112121211222121221222211111221221412412121111
14121211211141122144222421222412111141212121144112221
1222142121111121122122124211224111122122412212221242111

#32 (mean 1.583, $|D_3|=0.239$):
1111111111111111111111111111121112412211421214211
24112224121214442122121112112212121121222122121221212
122111211111412142142122142242221212411122121211412212
221111121412211211141444114212212411221112112221224111

#33 (mean 1.579, $|D_3|=0.218$):
42124122212121212112121212221112124111111222121212241
22211122111121412212422212121121141112221221122111
111112122111112211221222124411141242121221221422112
12122121222212112211221114212222111124222111111124412

#34 (mean 1.587, $|D_3|=0.217$):
1141121111111111112112144142414112121122111222112211122
12121222222212212221221221221221221111112111242122122121

1124121221212222412242121111111214111221242211211212211412
211221211212111211141111111111111411421411214222211121421121

#35 (mean 1.592, $|D_3|=0.214$):
111111111111111114122211121112121421442122121242212121241214
112112122421122111121411241222212242111221121222241111212122
1111112121121214121221211241241222211111411221112122241221
2221112221111121112111211224121122211111212411112111211111

#36 (mean 1.592, $|D_3|=0.216$):
11111111111212122221212111112121221142214112122222121121211
11212121221211111121141211421211122112114111114111121124211
111142112211422122114111411224211112124141221422221124214121
212411111122112121111211411112122122122211122411212211212211

#37 (mean 1.579, $|D_3|=0.220$):
22222224111111121211244221212111112111111114121212411224
1242111112111221212111422111112121111111112211241112212
1122112142111212242111221211111111121241241111142411211111
222121112242112241221111222212411112121112241211221111211121

#38 (mean 1.592, $|D_3|=0.225$):
11111111111124211211212112221221211112411212111121112411112
1212221224422112221142422112111122212114111121221211122142
14121444211112412422412112121112142121411211122111111241121
2211114111211121112111211112111111221122114222111111121214112

#39 (mean 1.587, $|D_3|=0.221$):
22411121222114111111224122211141121111111411212111241211211
111211122221214111142211411221122112212111212212111112111111
2414112111212421212111422221111412112112111121212121111111
144221211112124122122111222111212111422212121111224442112211

#40 (mean 1.579, $|D_3|=0.218$):
222222222214112211111121411121111211112111122241122111
112121411111224111211112222122112211122112112212111412142212
222122122221222112111121114112221121211112141111221122211411
121111211212411412212212212121141122144121111111242111212

#41 (mean 1.592, $|D_3|=0.225$):
1111111111111111111111111111111112211211111122141221412211
21111222111214122112211112221111112211121121421241424241211
4221211111411111141112411121212122111222122214122221221212
242121224111421442212111112221112124111141212242111112412212

#42 (mean 1.579, $|D_3|=0.238$):
22222222222222222222222222111221114112111111141124121111
1212411124111221121211111112111211112221111211221212121
2111141211111221111221221212121112121112121111121221111
21224111241112212221221111221112422211411212121224222212441

#43 (mean 1.579, $|D_3|=0.215$):
2224222422212122421121414121121112441121211112111124111222
111112221111222412211111111212121142112111121212224214221
221111212111221221121211112111121221112221221121111122214
21224121121111122121211112112114221111114211111212221412121

#44 (mean 1.592, $|D_3|=0.215$):
111111111111111112111221221122111142111221111224211121211411
42221111212111122244211112211112122412111212221412211422211
1122142112211112111122122411212112212211212212111211121214
11212111121411112222114122221121212221224411442221222112112

#45 (mean 1.587, $|D_3|=0.217$):
1211211224212211141411111111121212241122111111111211112111
122112222211211212224111241141222241242141211221121221421412
11111112221221211142211112122111211211412121111222222211412
11111112212121111121214111241111411121222112142211111141

#46 (mean 1.579, $|D_3|=0.215$):
222222222222222222222212111121111211214221114411221111112
12411111111212111112111222211112122121121111221111224122114
2111211444111114212211122221111211111212212121112222211111
214112141241214112114211112121211121142112111112111121212

#47 (mean 1.579, $|D_3|=0.229$):
22422222222222222222242224422222212211112411114411421111111
11112121121111124141211111212212121111222112121121122211121
11141112121112211111221212222112211212111111111212211111
12142121122212242111222212222122111111111111121112121121411

#48 (mean 1.579, $|D_3|=0.215$):
242222224212211121111111121211124411411111122422121421112
211121222212221241112112212212212112121221212421111141111211
1214141111111111211141121121222214211112122112122212111221
242421221211111111111222211122211211114112111112112212112

#49 (mean 1.592, $|D_3|=0.215$):
111111111141121142222122114411211221421214242121121244211111
212221212121121221421221141121211121221212112222221212111211
2211112121221111111212124111211221111221212111212111122
2222111211412122221144214212221111124111122224121212111111

#50 (mean 1.592, $|D_3|=0.216$):
12211212142111124212112211212121122121222142142111112111121
1412211111414211112221221214121221111144112111111111121221
211144121112111211212211221122221211412121111111121114111212
121411242124212212122122124112211114121112112122421111221211

#51 (mean 1.579, $|D_3|=0.229$):
222222222122124211122412221122112112111221122214122111112112
21112441121112142111211141111221111222442111111121122211121
2121221211111222221122121141221222112122111211211122112221
11111212222114111411212111121411222121121221122111212112

#52 (mean 1.579, $|D_3|=0.248$):
2224222222222222222222222222222222211211121111112121
21112221111112111241111122221211111122211121121122111211
2421111112211212242122211244211241412111122111221112111111
411112212211121211111241122112112114411211121121121221121

#53 (mean 1.592, $|D_3|=0.218$):
1111111111111111114212111442211121112121121142122141111121111
41212212142222222121111212121222221411122121211141122212
214411114241111121422111221212111212111212121241111222111212
1121221211222112122422212122114111211122122112221222111211

#54 (mean 1.579, $|D_3|=0.215$):
24222224422222442222222224211111212221111112111111111142
21121111212222111212111114211222114111112112212211211122111
2112421112212111111212111421211111412121212111121111221122
11114422122121114221111211211122211111221121114121111211421

#55 (mean 1.592, $|D_3|=0.233$):
11111111111111211441111414121422112121112211112122111211111
22112121111122212111112211211214211122412111122212211141112
211112142112121241112214224112411221241211112112112111212112
221141111242241212141211111411211424222121221141114111121121

#56 (mean 1.592, $|D_3|=0.279$):
114111242124112
241122111111211211421111141212242211112212124421112111141
442122221411121121122112142112422411114211214211212411112411
211111122212122122141211111221214412211212211441411111211142

#57 (mean 1.579, $|D_3|=0.215$):
2222222221121411221121122111111211141122212224211211121111
112114211112222211212412221221411211212411111211121121241222
2141221121121411111111214211242111112212141112122214121211
1221121112214221211114222212421122112112111211121211111111

#58 (mean 1.579, $|D_3|=0.218$):
22222222222211111212111444121114121122121211112112211241244
12211141111211211211111111212221214212211112212111441112
2211211122111242211211224122212111111142121112421112412112
121121122121112111122121211222112211122121112121122112221111

#59 (mean 1.579, $|D_3|=0.238$):
2222222222222222222222222222222222242222211221111112211
11211212211211214441121121111412124111211221211121121212111
1124211142114222112111221221111211241112121141211111121212
11111212211111211114211112211121112111111421121112112121211

#60 (mean 1.579, $|D_3|=0.214$):
222422222222222211221112421112111241212211142111211211141221
11112112121411221111111224121111421214114111122111141112
1121121221141411211144122121211121211221112421222111122221
111242112112111121111211412111112211112221122122221211

#61 (mean 1.579, $|D_3|=0.238$):
22222222222222224221111111121141111212112221114242121111111
12112211212421141422111122221112211111112122111111212221
212221421211112212221122122111211111112214221112211112222
212122221211211121142212212144122111212111221114211212112

#62 (mean 1.583, $|D_3|=0.230$):
1111111111111111111111111124122111112114212112242141411212
1112212221221411111211114222114121141114112121112121122241
11221141241111211122421141212112121224412242111141212111411
1221221111221124112112111122111222121121211212212221114122

#63 (mean 1.579, $|D_3|=0.222$):
2222222222222222222211412141121122211221111211112241112211
111221111121212214142211122111411211241112112211112142112211

21211421411111112121111211114111421422221112111121221241111
2221224111121121221111111112212111211122212211222121112211

#64 (mean 1.579, $|D_3|=0.215$):
222222222222222222221111141111214121141111121212241211112
2111211211112142111111412221221211114411111114112111242112
2111114122112211411222112112121411142112121221111221212211
22121211211211112111114112422112211212211121422111212221211

#65 (mean 1.579, $|D_3|=0.222$):
22221222112121111114121121411112221211111212112122122211
1112424112421211122111212111111221112112122222112111212211
11211212211221221221111221121222214112111212114122121211211
1421214211141122122222212111241121121111414221121111441221

#66 (mean 1.592, $|D_3|=0.216$):
1121212122111111
2211222124122421122111211121111221222411142221212112224211
1241124112114411211211122242141211411122112222212121212122
2222112121142412124424211211121214211124211242221212112211

#67 (mean 1.592, $|D_3|=0.215$):
111112112111411111122221212111411141112112111121212411412221
12222211121221411411221212221211211411212112411211122112212
2141112141112212211121411121111212112121221224121411122211121
412111442112121112121212412211212122212111212122211222111211222

#68 (mean 1.579, $|D_3|=0.216$):
2222222222222424242111212111121241221144212112114111111221
21211212111111244212112212222211212111211121211211141211112
11212111412212111141211211111122122111221111112112411212
122121111141111114411221111111224212112124111222111112111

#69 (mean 1.592, $|D_3|=0.216$):
1111112122411112212111141111122121121224112221222221142111
111114221211242212141122224242211214242111221111221211222111
141111121212122112211111212211122114111112422121112111111
12124112221121411111411122421121112111412121121111421212

#70 (mean 1.592, $|D_3|=0.216$):
211142121122412224111224112112121121211211111412112222
1122111212121412122121211111122411212121211112211121212424
1122211212121212114221122211121121141114222111222411111111
124122121114121111221212211211121442111114422412111111112

#71 (mean 1.592, $|D_3|=0.214$):
111111111111111122121412142111212211114122144221122222
1144144124242112221411121121212111112111421222112221212211
12222421122121112122221241111411212211121121112121111222
2112122112142121211112111221121241211211122141211121111112

#72 (mean 1.579, $|D_3|=0.214$):
2422422221221111141112111241121121212211221111221212121
224411114221122111224111111212121111211212114212121141242111
2121112112221122214122411112111121221422121212122111122121
1222122121212111121112141121112111224111111121114111114

#73 (mean 1.587, $|D_3|=0.221$):
1211411111224111211211242211212122111122211112111124214211
1222211111112212111212141211144111221211112121111221212121
112211111112412421211222122111141211241111111211121121241
21112224221411112241121121112122214121212224111122111411241

#74 (mean 1.592, $|D_3|=0.215$):
1111111111111111121221121214222221141111221212112121122111
111221221141242112222111122122121211212224121122221142212212
212221221112121111122242122111111121212121211112114244142
2214112111241112114111211111122111141224221222211411211124

#75 (mean 1.579, $|D_3|=0.215$):
2211241114212121212111222211111124212112121211141221211
212111412114111121211112114112124121124111121121221221224412
2211112222111121221221111112121212222114111211422221111211
114211221121212122121112122124124112121112122112111114121

#76 (mean 1.579, $|D_3|=0.225$):
2222242111221411122111112421111112111211212411111214111112
4211112111212121211111224121112142121211121111112221122122
2111221212212111411211124221242111241122112111111111212414
211141212112111121111212221214142211112112211114412214421112

#77 (mean 1.583, $|D_3|=0.241$):
1111111122221112111121121111222121141412224214222221242142
22112211412422121212211211112412111212121112111112122221
1211221121211422221114222211112411112121112211111221221122
1212212211242121112121122122121112212211122121111221212112

#78 (mean 1.592, $|D_3|=0.215$):
1111111111121121112122211442211222212222211112112212111412
1122211222121142414111122212111121114111212441111211222211
2212221211212111211222144122111111141212222111121222212222
1111224121112211121112112111422122214214111122111114411211

#79 (mean 1.587, $|D_3|=0.221$):
141211221111114212212121112121122221112111111114111111124
112114142221221111211221421222212111211221111114414124222
111111241121411121112211111124211412111142111221211122112112
124112141112221112212122112111241141112121221221121111222242

#80 (mean 1.587, $|D_3|=0.217$):
11111214421111221114121211211121122211112212121112122111121
21122111112141421221211142214221211121211121122112121122142
12111221221212221141222121211121111122124111114211122242
12211224222111241114212121211211242421212111221112111121112

#81 (mean 1.592, $|D_3|=0.225$):
111111111111111111241411112222112141212121111412114211421
112212111111211211212121111122424111111221142112111212141
22221121112114221112224111242221111114122112122124122211112
111112121112221211224114211124221242122214211111121412211121

#82 (mean 1.583, $|D_3|=0.214$):
21212124241211112141121211212211211212121212422111111121214
22411121112112441222112212211122212121122141142214112424421
124211221112111112112224111121121212111221112111112112212121
1111111112221221112141122111211112221111122112241211221221111

#83 (mean 1.592, $|D_3|=0.215$):
11221211112421211221121111211111411221212122211111111122
221111224111222212221222121111212222121212111422421441221212
2411122212212111214141224422211122411112112121221112111421
4111121211211211414111122121221411211212111112221112121111

#84 (mean 1.592, $|D_3|=0.218$):
122111111121141121122212142221142212112221111212122121211211
12121111211212242141111141121242211112122221111211144112242
1211111111221211221222221221144122111112122212211141111214
2111212142211421112121112111211411211122212112222221112221

#85 (mean 1.579, $|D_3|=0.222$):
21212244112222114222122121211112111121114141141211111122
12212111112112121221211211211212211211412221211211121122221211
22212421221121112112112412121411112212411112211221211112212
44122122121211112212122411121212122121222111211412121211111

#86 (mean 1.579, $|D_3|=0.238$):
2222222224222422222222221221111212241212211121242112214
112111111111112112214211111122111211111212121412221111142
2112211111122111121112121112141221212121211122121221122112
212122221111411424212211212121111222221212121114111112112

#87 (mean 1.579, $|D_3|=0.229$):
422222222222222422422214222211122212121112111211122111412
11122111211111422111221111121124112111221224111111121111
11212221111111111114111122411121122121111112211211242222
221122221111112111211241421121221122122242211112121212211221

#88 (mean 1.587, $|D_3|=0.227$):
2421412121212121214121111411122212121111212211212121212222
121112221111211111212412111211122122111111121112112122121
1212421224122121112121222111112212122221212421212121221122
11222421112222121244222111121214121111121142211111124122

#89 (mean 1.579, $|D_3|=0.218$):
42222222221141112112211121212122111111212111111122122121
121122211212212221111241211211141121142111121121211121114
2412211111221114111112241411212111211124222122222212122242
2112211112112112211111122212111121224111421411124122221112

#90 (mean 1.592, $|D_3|=0.216$):
1111111121121114212212121112122112211121211112142211142111
2111221111244121112222142121111112121121121142211112421112
1111212122122221212241421122111212121212121112111411421211
1111211411122121211112222122141141111224414212111241224111

#91 (mean 1.579, $|D_3|=0.216$):
42222242222121411121211212121411121111411121211121111211112
11211212122124112111214122121211111111112112224111211122212
1244122141421144121211211111241221112111122221221212111211
2212211111124122111111121112211211112212114121114214121222

#92 (mean 1.583, $|D_3|=0.225$):
111111212222111221221141121122112242122221111124221211112112
1241112224112112411211221121111111212241224221111111112212

2111211122121211121121111221211121212224222222211122121212
11211211222124124242114121111211111221111121242111241211221

#93 (mean 1.592, $|D_3|=0.220$):
111111111111111111122124212111114141212221211112112212111
112214111121221111122212412121111241211212121221212112122
442421122114211241121211112211211121214111244114111124122
221112111211222122121211424212242121411212111411122112122111

#94 (mean 1.579, $|D_3|=0.214$):
4422222222222242412111111111121112211121124142112211211
11211212212122421212241121111221112212421211211141112111211
1412421121111212211122241141211112412211111111142111112
22211122111112121112142111212221221112112111121412212112211

#95 (mean 1.579, $|D_3|=0.238$):
22222222222422222242222222222221121111111112212121
24111111211112112211124112112111121421211211212111211221
11124122221111141112212211111422241211211121212111111112
211112212221114111221222111124224111111112111222412211111

#96 (mean 1.579, $|D_3|=0.238$):
222222242222222222212112242111121111212112124121211122122
2211222212111121114122111111121412211122112121221111221112
12111412211411112211121221111222112124141211212124221211141
12221211111121121241221111222212112111111221121112421122121

#97 (mean 1.592, $|D_3|=0.233$):
1111111111111111111211414222111121122111222211112111142412
211211421421114212121224111422124111212421114221122212211112
11221121111112121212221112111412121122221412141111211221
121411112221411111222112211141121141211241121224411112411

#98 (mean 1.579, $|D_3|=0.229$):
222222222222222222211141211111221112212221111112121221
121211142212111221111412221221111221112111121112121111122
111212241411111111412222221211112111121212112222114111124
1241212224112112111124224122121142112111221112241121112212

#99 (mean 1.592, $|D_3|=0.218$):
1111111111111111141221141424224211111241241111221121112212
12211114414121112211112412212114112111112211212442221224221
1222121222121211111121122224214122211221112221122111212212
1221121111212222121222211122111121112222141111121121212211

#100 (mean 1.579, $|D_3|=0.215$):
22222221111441121124221121114121221122212141112121111121221
2141122121222122212221222111212142112212211124411411211111
1144111141121114114111114211221122112111211121212121212211
21211211111121221211111122211222122122111211121121411211

#101 (mean 1.579, $|D_3|=0.218$):
2222222222221111212212111121114212112121111221141241211
12111411111211212211111121212411222211222112121111211122111
14222211211211141112111212124212222111142422212211421221
2224111211114121412121111211112111121212112111424122111

#102 (mean 1.579, $|D_3|=0.218$):
222222222222222222222242222222222121211224211222122412112
21212111121121211122122114112111114112421111112111121114111
121122111112111112111142112111421111111111224111244421121
1122111111121112112141111121111242411221111122211212112121

#103 (mean 1.579, $|D_3|=0.229$):
2222222222121111211121112241122221121211122122111221111122
1221111212421121121211121241212121121222442121112121442211
211212121111112422121121212112422111221111122211112121112
121412111211212211222111241222124121114211111121241121112

#104 (mean 1.587, $|D_3|=0.217$):
121142221212121221211142111221111221221122214121212122221
1111214112111211221141111122212111122111112111122122222
2114212112111121211112141114214112122211122111121421241221
21111121412221211412121122211211114221141122121221224211141

#105 (mean 1.592, $|D_3|=0.229$):
11111111111111111111111111212122111112122112211211411221
1121211111111211221112221222421112142112421222112122242222
1124212121111121222112244111222242122211211122212422122241
2221122411211212411122121212124211222111211211121241122212

#106 (mean 1.583, $|D_3|=0.232$):
111111211422221221211124111112112224114221122122111422111
211122221211122112221222121121111222212111212121242221122
2121221111121122111122121111222114224211214221122112412142
122141114121211211222111112111122421112212111122111221112

#107 (mean 1.579, $|D_3|=0.218$):
2422222222221221111111112221412122211111221122111111111111
12121111111142121221111142222411212111211111111121212221
2221212221112111212142211112121211412142112111221212212211
1121111144122122111142222121412222111242221141222211111424

#108 (mean 1.579, $|D_3|=0.214$):
2422222222222212122121122111212112212211211112224214414122
1121111112111141121111222111111221421112211111142211122
11111122222222122211414111141421112121111111212112121411212
1111221414411122211112121112111112421122221111121211141

#109 (mean 1.592, $|D_3|=0.220$):
1111111111112242124214214112211121421222121122111242111214
2111212111112214114142121112124211111111221221111242142121
112142212222111121111124124221111111212112111212211211212
412111112211112111211111121211142422211412242122221112122

#110 (mean 1.592, $|D_3|=0.220$):
11111111112221421121222122121111141121421112412411211142121
2111112111111112411212121111222221241121111422221221122
24222221111121142121111111211121122442111211112121124211
1122122111211111214111221421142122122124211421214112111214

#111 (mean 1.579, $|D_3|=0.229$):
2222221212212112111212121112111111211421222122111141212
124112121221211111221212211241111222122111211211121222111
122414111111221124441212212121111211224212222112122112221
11212122111221121212121212124112121121242111211221422421

#112 (mean 1.583, $|D_3|=0.218$):
111111111121211121141221221412211122211111211141111122111
241121421211112121212411422112422111121112222144141214222121
112121212221112221121112121111112122111112112111121112121
21244211141112221212211222112221112442422111211144221211111

#113 (mean 1.579, $|D_3|=0.248$):
222242222222222222112122211122221112111142211121212111241
1111122121212122421111212221112211211221212111121221111111
112222112211112112122211211114221142121221121414111221112
2124112121211112112211211221412222112142221212142221111112

#114 (mean 1.579, $|D_3|=0.229$):
2222222222222222221111121121112121212124111122411112222
11222212211112111121121211141122211412411111222112411211
211214212111111111441211222122111121211421241121211221222
14221212112222121222112121121211141112112111211112214211

#115 (mean 1.592, $|D_3|=0.220$):
111111111111112121121114222221221222242212211211114111
222212121121111111114411422121212141112422141242112112112
12121141112122122211111241214422111222122121114121121241211
12221144111211212111211111121111421111222411422211211111

#116 (mean 1.592, $|D_3|=0.233$):
112112112114212221244
112111422124111121241222121142222121242122112121222111222212
22141441211411111112121242114121421412124212211111211112222
1121111111211412112111241111224222141121222124124111121111

#117 (mean 1.579, $|D_3|=0.222$):
22222222112222111141111122214211221212422112211111122241221
211211211211222111211121211112111121112221224211111121211422
11111121111212224111112112112211111211121421111221121222
412222112212141111121412222211111212114124211241222114112221

#118 (mean 1.587, $|D_3|=0.214$):
12412224222214111212111112121211212211112121112111121422111
2112211111211111221141112221111222121111211121111242212111
11122112421112211114112111212211112142112242211122221224144
12422122221111421211121111242411212222111112112122121142

#119 (mean 1.579, $|D_3|=0.218$):
22224222222222222222222222222242112112112141111111221142
111212111211121222122112111221211122111211211111121124111
11121112211114211111121111112414212242214111222212111122
14112122111111121121211412112211111211211111141422241211

#120 (mean 1.579, $|D_3|=0.214$):
222224222222222222122411212122111121111111121211211212111
21221112112111211211111121141221122111211221112224141112111
12112212121122114121212114122111114111221211111211422444
41121111112121122241141221241241112121121211111121121211221